

Deep Residual Learning and Convolutional Neural Networks for Image Recognition

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ABSTRACT

Deeper convolutional neural networks are notoriously difficult to train due to vanishing and exploding gradient problems. We present a deep residual learning framework where layers explicitly learn residual mapping with reference to layer inputs. On the ImageNet dataset, deep residual networks achieve state-of-the-art image classification accuracy, confirming that identity skip connections facilitate gradient propagation throughout ultra-deep architectures.

Keywords: Deep Learning, Convolutional Neural Networks, Residual Networks, Computer Vision, Image Classification

1. Introduction

Deep convolutional neural networks have led to breakthroughs for image classification. However, as networks grow deeper, accuracy becomes saturated and degrades rapidly. This degradation indicates that not all systems are similarly easy to optimize.

2. Residual Formulation

Instead of hoping every few stacked layers directly fit a desired underlying mapping $H(x)$, we let these layers approximate a residual mapping $F(x) := H(x) - x$. The original mapping is recast into $F(x) + x$, realized by feedforward networks with shortcut connections.

3. Experimental Findings

Residual networks with depths of up to 152 layers were evaluated on ImageNet. The networks converged significantly faster and attained lower top-1 error rates without degradation.

4. Conclusion

Residual learning represents a fundamental breakthrough in training deep neural networks for visual recognition, object detection, and semantic segmentation.